

hw3

AUTHOR

Andre Gala-Garza

PUBLISHED

March 27, 2026

Disclaimer: Generative AI was used to assist with templating and writing code in this assignment; however, this code was checked manually and edited by hand to ensure accuracy.

Source: OpenAI. (2026). *ChatGPT* [Large language model]. <https://chatgpt.com/>.

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
```

```
✓ dplyr      1.2.0    ✓ readr      2.1.5
```

```
✓ forcats   1.0.1    ✓ stringr    1.5.1
```

```
✓ ggplot2   4.0.1    ✓ tibble     3.3.0
```

```
✓ lubridate 1.9.4    ✓ tidyr      1.3.1
```

```
✓ purrr     1.1.0
```

```
— Conflicts — tidyverse_conflicts() —
```

```
✗ dplyr::filter() masks stats::filter()
```

```
✗ dplyr::lag()     masks stats::lag()
```

```
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidytext)
```

```
library(dplyr)
```

```
library(textdata)
```

Exploratory Data Analysis

```
pubmed <- read.csv("pubmed.csv")
```

```
colnames(pubmed)
```

```
[1] "abstract" "term"
```

```
head(pubmed$term)
```

```
[1] "covid" "covid" "covid" "covid" "covid" "covid"
```

```
unique(pubmed$term)
```

```
[1] "covid" "meningitis" "prostate cancer" "cystic fibrosis"
```

```
[5] "preeclampsia"
```

```
head(pubmed$abstract)
```

[1] "Background and aims: Many patients with coronavirus disease 2019 (COVID-19) have underlying cardiovascular (CV) disease or develop acute cardiac injury during the course of the illness. Adequate understanding of the interplay between COVID-19 and CV disease is required for optimum management of these patients. Methods: A literature search was done using PubMed and Google search engines to prepare a narrative review on this topic. Results: Respiratory illness is the dominant clinical manifestation of COVID-19; CV involvement occurs much less commonly. Acute cardiac injury, defined as significant elevation of cardiac troponins, is the most commonly reported cardiac abnormality in COVID-19. It occurs in approximately 8-12% of all patients. Direct myocardial injury due to viral involvement of cardiomyocytes and the effect of systemic inflammation appear to be the most common mechanisms responsible for cardiac injury. The information about other CV manifestations in COVID-19 is very limited at present. Nonetheless, it has been consistently shown that the presence of pre-existing CV disease and/or development of acute cardiac injury are associated with significantly worse outcome in these patients. Conclusions: Most of the current reports on COVID-19 have only briefly described CV manifestations in these patients. Given the enormous burden posed by this illness and the significant adverse prognostic impact of cardiac involvement, further research is required to understand the incidence, mechanisms, clinical presentation and outcomes of various CV manifestations in COVID-19 patients."

[2] "Introduction: Contradictory data have been reported on the incidence of stroke in patients with COVID-19 and the risk of SARS-CoV-2 infection among patients with history of stroke. Methods: This study systematically reviews case series reporting stroke as a complication of COVID-19, and analyses the prognosis of patients with COVID-19 and history of stroke. The pathophysiological mechanisms of stroke in patients with COVID-19 are also reviewed. Conclusions: History of stroke increases the risk of death due to COVID-19 by 3 times. Stroke currently seems not to be one of the main complications of COVID-19."

[3] "This article aims at collecting all information needed for dentists regarding the COVID-19 pandemic throughout the world by reviewing articles published by now. In late 2019, a pneumonia outbreak of uncertain etiology happened in Wuhan, China. There were many reports related to a live-animal and seafood market, supporting that the pathogens were transferred from animals to humans, rapidly evolving into transmission from human to human. The pathogen was classified as 2019 Novel Corona Virus (2019-nCoV), and the disease was named Corona Virus Disease 2019 (COVID-19). Given that COVID-19 has lately been detected in infected patients' saliva, the COVID-19 outbreak is an alert that all dental and other health professionals must be vigilant in defending against the infectious disease spread, and it may enable to assess whether non-invasive saliva diagnostic for COVID-19. There has so far been no evidence from randomized controlled trials to prescribe any particular anti-nCoV treatment or vaccine, and COVID-19 management has been widely supportive. Since the ACE-2 was expressing on oral cavity mucosa, there is a potentially huge COVID-19 infectious vulnerability risk for oral cavity and brought up a proof for the future prevention procedure in dental practice and daily life. As a result, the whole dental teams should be vigilant and keep patients and themselves in a safe environment by following the guideline in this study."

[4] "OBJECTIVE. The objective of our study was to determine the misdiagnosis rate of radiologists for coronavirus disease 2019 (COVID-19) and evaluate the performance of chest CT in the diagnosis and management of COVID-19. The CT features of COVID-19 are reported and compared with the CT features of other viruses to familiarize radiologists with possible CT patterns. MATERIALS AND METHODS. This study included the first 51 patients with a diagnosis of COVID-19 infection confirmed by nucleic acid testing (23 women and 28 men; age range, 26-83 years) and two patients with adenovirus (one woman and one man; ages, 58 and 66 years). We reviewed the clinical

information, CT images, and corresponding image reports of these 53 patients. The CT images included images from 99 chest CT examinations, including initial and follow-up CT studies. We compared the image reports of the initial CT study with the laboratory test results and identified CT patterns suggestive of viral infection. RESULTS. COVID-19 was misdiagnosed as a common infection at the initial CT study in two inpatients with underlying disease and COVID-19. Viral pneumonia was correctly diagnosed at the initial CT study in the remaining 49 patients with COVID-19 and two patients with adenovirus. These patients were isolated and obtained treatment. Ground-glass opacities (GGOs) and consolidation with or without vascular enlargement, interlobular septal thickening, and air bronchogram sign are common CT features of COVID-19. The "reversed halo" sign and pulmonary nodules with a halo sign are uncommon CT features. The CT findings of COVID-19 overlap with the CT findings of adenovirus infection. There are differences as well as similarities in the CT features of COVID-19 compared with those of the severe acute respiratory syndrome. CONCLUSION. We found that chest CT had a low rate of missed diagnosis of COVID-19 (3.9%, 2/51) and may be useful as a standard method for the rapid diagnosis of COVID-19 to optimize the management of patients. However, CT is still limited for identifying specific viruses and distinguishing between viruses."

[5] "Background: Much of the focus regarding the global pandemic of coronavirus disease of 2019 (COVID-19) has been on the cardiovascular, pulmonary, and hematologic complications. However, neurologic complications have arisen as an increasingly recognized area of morbidity and mortality. Objective: This brief report summarizes the neurologic complications associated with COVID-19 with an emphasis on the emergency medicine clinician. Discussion: COVID-19 has infected over 3.5 million people and killed over 240,000 people worldwide. While pulmonary complications are profound, the neurologic system is also significantly impacted, with complications including acute cerebrovascular events, encephalitis, Guillain-Barré syndrome, acute necrotizing hemorrhagic encephalopathy, and hemophagocytic lymphohistiocytosis. Additionally, patients on immunosuppressive medications for pre-existing neurologic issues are at an increased risk for complications with COVID-19 infection, and many of the currently proposed COVID-19 therapies can interact with these medications. Conclusions: When caring for COVID-19 patients, emergency medicine clinicians should be aware of the neurologic complications from COVID-19."

[6] "The World Health Organization (WHO) has issued a warning that, although the 2019 novel coronavirus (COVID-19) from Wuhan City (China), is not pandemic, it should be contained to prevent the global spread. The COVID-19 virus was known earlier as 2019-nCoV. As of 12 February 2020, WHO reported 45,171 cases and 1115 deaths related to COVID-19. COVID-19 is similar to Severe Acute Respiratory Syndrome coronavirus (SARS-CoV) virus in its pathogenicity, clinical spectrum, and epidemiology. Comparison of the genome sequences of COVID-19, SARS-CoV, and Middle East Respiratory Syndrome coronavirus (MERS-CoV) showed that COVID-19 has a better sequence identity with SARS-CoV compared to MERS CoV. However, the amino acid sequence of COVID-19 differs from other coronaviruses specifically in the regions of 1ab polyprotein and surface glycoprotein or S-protein. Although several animals have been speculated to be a reservoir for COVID-19, no animal reservoir has been already confirmed. COVID-19 causes COVID-19 disease that has similar symptoms as SARS-CoV. Studies suggest that the human receptor for COVID-19 may be angiotensin-converting enzyme 2 (ACE2) receptor similar to that of SARS-CoV. The nucleocapsid (N) protein of COVID-19 has nearly 90% amino acid sequence identity with SARS-CoV. The N protein antibodies of SARS-CoV may cross react with COVID-19 but may not provide cross-immunity. In a similar fashion to SARS-CoV, the N protein of COVID-19 may play an important role in suppressing the RNA interference (RNAi) to overcome the host defense. This mini-review aims at investigating the most recent trend of COVID-19."

Text Mining

Problem 1

```
# Number of articles per search term
pubmed %>%
  count(term, sort = TRUE)
```

	term	n
1	covid	981
2	prostate cancer	787
3	preeclampsia	780
4	cystic fibrosis	376
5	meningitis	317

From this frequency table, we see that "covid" is the search term with the most articles, followed by "prostate cancer" and "preeclampsia". Meanwhile, "cystic fibrosis" and "meningitis" have significantly fewer search results.

```
# Most common tokens overall
pubmed %>%
  unnest_tokens(token, abstract) %>%
  count(token, sort = TRUE) %>%
  head(10)
```

	token	n
1	the	28126
2	of	24760
3	and	19993
4	in	14653
5	to	10920
6	a	8245
7	with	8038
8	covid	7275
9	19	7080
10	is	5649

From this table, we see that nearly all of the most common tokens are stopwords like "and", "the", or "of", which do not provide significant meaning. The most common tokens that are not stopwords are "covid" and "19", highlighting the impact of the COVID-19 pandemic on the content of PubMed articles.

```
# 5 most common tokens per search term
pubmed %>%
  unnest_tokens(token, abstract, drop = FALSE) %>%
  count(term, token, sort = TRUE) %>%
  group_by(term) %>%
  slice_max(n, n = 5, with_ties = FALSE) %>%
  ungroup()
```

```
# A tibble: 25 × 3
  term      token      n
  <chr>    <chr>    <int>
1 covid    the      9741
2 covid    of       7668
3 covid    covid    7275
4 covid    19       7035
5 covid    and      6861
6 cystic   the      2677
7 cystic   of       2304
8 cystic   and      1777
9 cystic   in       1497
10 cystic  to       1011
11 meningitis the      2489
12 meningitis of      2269
13 meningitis and     1683
14 meningitis in     1322
15 meningitis a       830
16 preeclampsia the    7119
17 preeclampsia of     6837
18 preeclampsia and    5255
19 preeclampsia in     3869
20 preeclampsia to     2651
21 prostate cancer the  6100
22 prostate cancer of   5682
23 prostate cancer and  4417
24 prostate cancer cancer 3840
25 prostate cancer prostate 3832
```

From viewing the 5 most common tokens per search term, we see that all of them are stopwords except for "covid" and "19" for the search term "covid", and "cancer" and "prostate" for the search term "prostate cancer".

Problem 2

```
# Most common tokens while removing stop words
pubmed %>%
  unnest_tokens(token, abstract) %>%
  anti_join(stop_words, by = c("token" = "word")) %>%
  count(token, sort = TRUE) %>%
  head(10)
```

```
      token      n
1      covid 7275
2        19 7080
3  patients 4674
4    cancer 3999
5  prostate 3832
```

```

6      disease 2574
7      pre 2165
8      eclampsia 2005
9      preeclampsia 1863
10     treatment 1841

```

After removing stopwords, we now find that the most common tokens are much more closely related to the search terms themselves, such as "covid", "cancer", "prostate", and "preeclampsia". We also see that "disease" and "treatment" are very common tokens, which make sense because those concepts can be applied to nearly any medical case.

Problem 3

```

# Tokenize abstracts into bigrams
bigrams <- pubmed %>%
  unnest_tokens(bigram, abstract, token = "ngrams", n = 2)

# 10 most common bigrams
top_bigrams <- bigrams %>%
  count(bigram, sort = TRUE) %>%
  head(10)

top_bigrams

```

```

      bigram      n
1      covid 19 6969
2      of the 3883
3      in the 3418
4 prostate cancer 3222
5    pre eclampsia 1854
6 patients with 1587
7      of covid 1519
8      and the 1154
9      to the 1061
10    of prostate 870

```

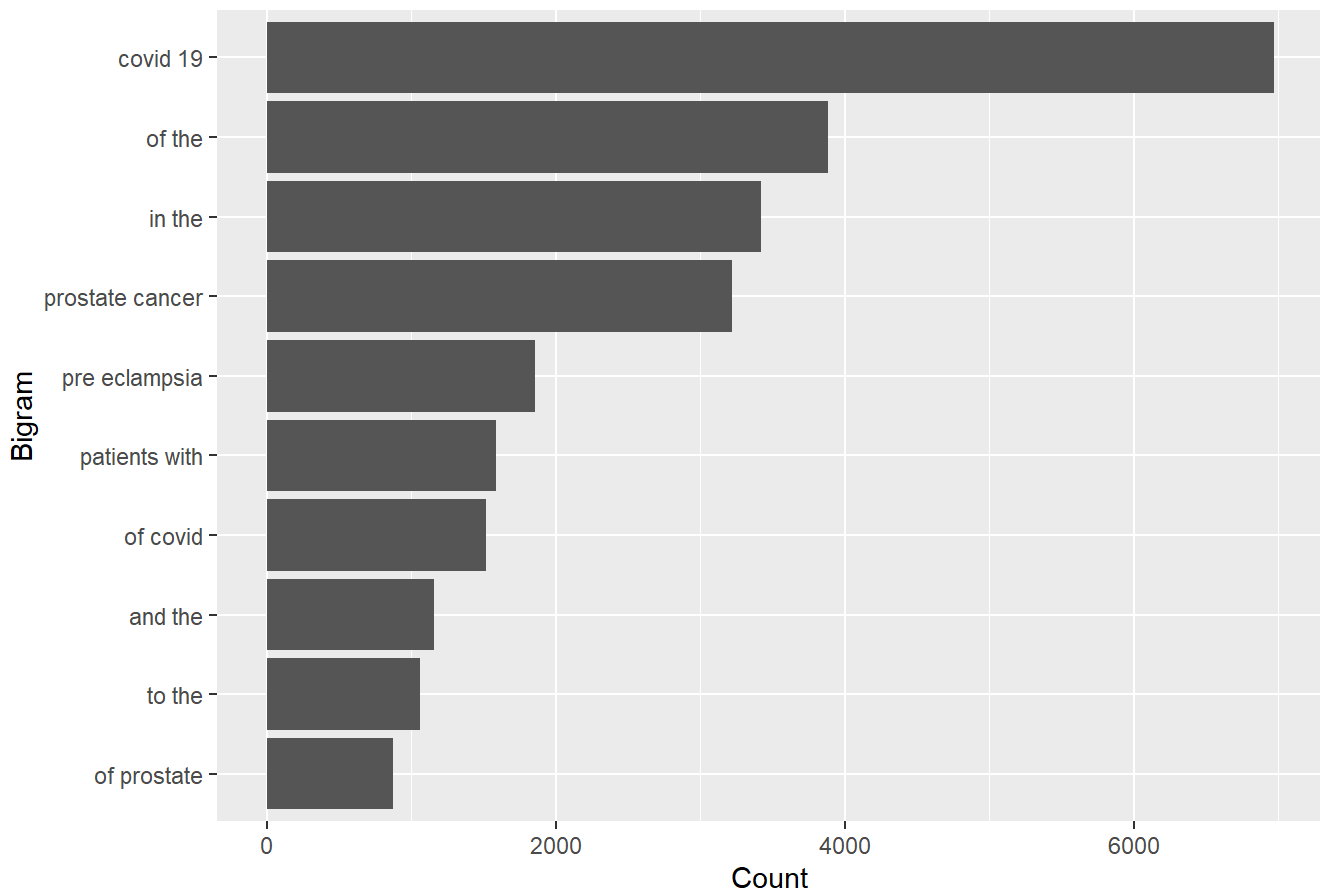
When including stopwords in the abstracts, we find that the most common bigram is "covid 19" (further demonstrating the impact of the COVID-19 pandemic on the content of abstracts), followed by two stopword bigrams, then "prostate cancer" and "pre eclampsia".

```

# Visualize the 10 most common bigrams
top_bigrams %>%
  ggplot(aes(x = reorder(bigram, n), y = n)) +
  geom_col() +
  coord_flip() +
  labs(x = "Bigram", y = "Count", title = "10 Most Common Bigrams")

```

10 Most Common Bigrams



Problem 4

```
# Calculate TF-IDF for each token within each search term
```

```
tf_idf_words <- pubmed %>%
  unnest_tokens(token, abstract) %>%
  anti_join(stop_words, by = c("token" = "word")) %>%
  count(term, token, sort = TRUE) %>%
  bind_tf_idf(token, term, n)
```

```
head(tf_idf_words)
```

	term	token	n	tf	idf	tf_idf
1	covid	covid	7275	0.06468852	1.609438	0.10411215
2	covid	19	7035	0.06255446	0.000000	0.00000000
3	prostate cancer	cancer	3840	0.05641169	0.000000	0.00000000
4	prostate cancer	prostate	3832	0.05629416	1.609438	0.09060196
5	covid patients	patients	2293	0.02038911	0.000000	0.00000000
6	preeclampsia	pre	2038	0.02594889	0.000000	0.00000000

```
# 5 tokens with the highest TF-IDF for each search term
```

```
tf_idf_words %>%
```

```
group_by(term) %>%
slice_max(tf_idf, n = 5, with_ties = FALSE) %>%
ungroup() %>%
arrange(term, desc(tf_idf))
```

A tibble: 25 × 6

	term	token	n	tf	idf	tf_idf
	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	covid	covid	7275	0.0647	1.61	0.104
2	covid	pandemic	800	0.00711	1.61	0.0114
3	covid	coronavirus	647	0.00575	1.61	0.00926
4	covid	sars	372	0.00331	1.61	0.00532
5	covid	cov	334	0.00297	1.61	0.00478
6	cystic fibrosis	cf	625	0.0234	0.916	0.0215
7	cystic fibrosis	fibrosis	867	0.0325	0.511	0.0166
8	cystic fibrosis	cystic	862	0.0323	0.511	0.0165
9	cystic fibrosis	cftr	86	0.00322	1.61	0.00518
10	cystic fibrosis	sweat	83	0.00311	1.61	0.00500
11	meningitis	meningitis	429	0.0164	1.61	0.0264
12	meningitis	meningeal	219	0.00837	1.61	0.0135
13	meningitis	pachymeningitis	149	0.00569	1.61	0.00916
14	meningitis	csf	206	0.00787	0.916	0.00721
15	meningitis	meninges	106	0.00405	1.61	0.00652
16	preeclampsia	eclampsia	2005	0.0255	1.61	0.0411
17	preeclampsia	preeclampsia	1863	0.0237	1.61	0.0382
18	preeclampsia	pregnancy	969	0.0123	0.511	0.00630
19	preeclampsia	maternal	797	0.0101	0.511	0.00518
20	preeclampsia	gestational	191	0.00243	1.61	0.00391
21	prostate cancer	prostate	3832	0.0563	1.61	0.0906
22	prostate cancer	androgen	305	0.00448	1.61	0.00721
23	prostate cancer	psa	282	0.00414	1.61	0.00667
24	prostate cancer	prostatectomy	215	0.00316	1.61	0.00508
25	prostate cancer	castration	148	0.00217	1.61	0.00350

The TF-IDF evaluates how important each word in the abstract is to the "document" (the search term). The results in Question 1 instead only tally the frequency of each word for each search term.

The Question 1 results are dominated by very common words such as "the," "of," "and," and "in"; these appear often in every set of abstracts, but do not tell us much about what makes each search term distinctive. By contrast, the TF-IDF results highlight words that are especially characteristic of each search term relative to the others: for example, "covid," "pandemic," and "coronavirus" stand out for the covid abstracts, while "cf," "cftr," and "sweat" are more specific to cystic fibrosis, and "meningitis," "meningeal," and "meninges" are more specific to meningitis. Similarly, "eclampsia" and "preeclampsia" stand out for preeclampsia, and "prostate," "psa," and "prostatectomy" stand out for prostate cancer. Overall, the Question 1 results shows the most repeated words, while TF-IDF is more useful for identifying the words that best distinguish one topic from another.

Sentiment Analysis

Problem 5

```
# Sentiment analysis with NRC lexicon
nrc_sentiments <- pubmed %>%
  unnest_tokens(token, abstract) %>%
  inner_join(get_sentiments("nrc"), by = c("token" = "word"))
```

Warning in inner_join(., get_sentiments("nrc"), by = c(token = "word")): Detected an unexpected many-to-many relationship between `x` and `y`.
 i Row 8 of `x` matches multiple rows in `y`.
 i Row 3753 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

```
# Most common sentiment for each search term
nrc_sentiments %>%
  count(term, sentiment, sort = TRUE) %>%
  group_by(term) %>%
  slice_max(n, n = 1, with_ties = FALSE) %>%
  ungroup()
```

A tibble: 5 × 3

	term	sentiment	n
	<chr>	<chr>	<int>
1	covid	positive	9874
2	cystic fibrosis	positive	2747
3	meningitis	negative	2111
4	preeclampsia	positive	8014
5	prostate cancer	negative	8918

With all NRC categories included, covid, cystic fibrosis, and preeclampsia are dominated by "positive," while meningitis and prostate cancer are dominated by "negative," which gives a broad sense of tone but not much detail.

```
# Most common sentiment for each search term,
# removing "positive" and "negative"
nrc_sentiments %>%
  filter(!sentiment %in% c("positive", "negative")) %>%
  count(term, sentiment, sort = TRUE) %>%
  group_by(term) %>%
  slice_max(n, n = 1, with_ties = FALSE) %>%
  ungroup()
```

A tibble: 5 × 3

	term	sentiment	n
	<chr>	<chr>	<int>
1	covid	fear	7730
2	cystic fibrosis	disgust	1714

3	meningitis	fear	1510
4	preeclampsia	anticipation	4780
5	prostate cancer	fear	8118

After removing those two general categories, more specific emotional patterns appear: covid, meningitis, and prostate cancer are most associated with "fear," which makes sense given their seriousness and risk; preeclampsia is most associated with "anticipation," possibly reflecting monitoring, prediction, and prevention in pregnancy care; and cystic fibrosis is most associated with "disgust," likely because of how certain disease-related terms are classified in the NRC lexicon rather than because the abstracts are written in an emotional style.

Problem 6

```
# Create an abstract ID
pubmed <- pubmed %>%
  mutate(abstract_id = row_number())

# Get average AFINN sentiment score for each abstract
afinn <- lexicon_afinn()

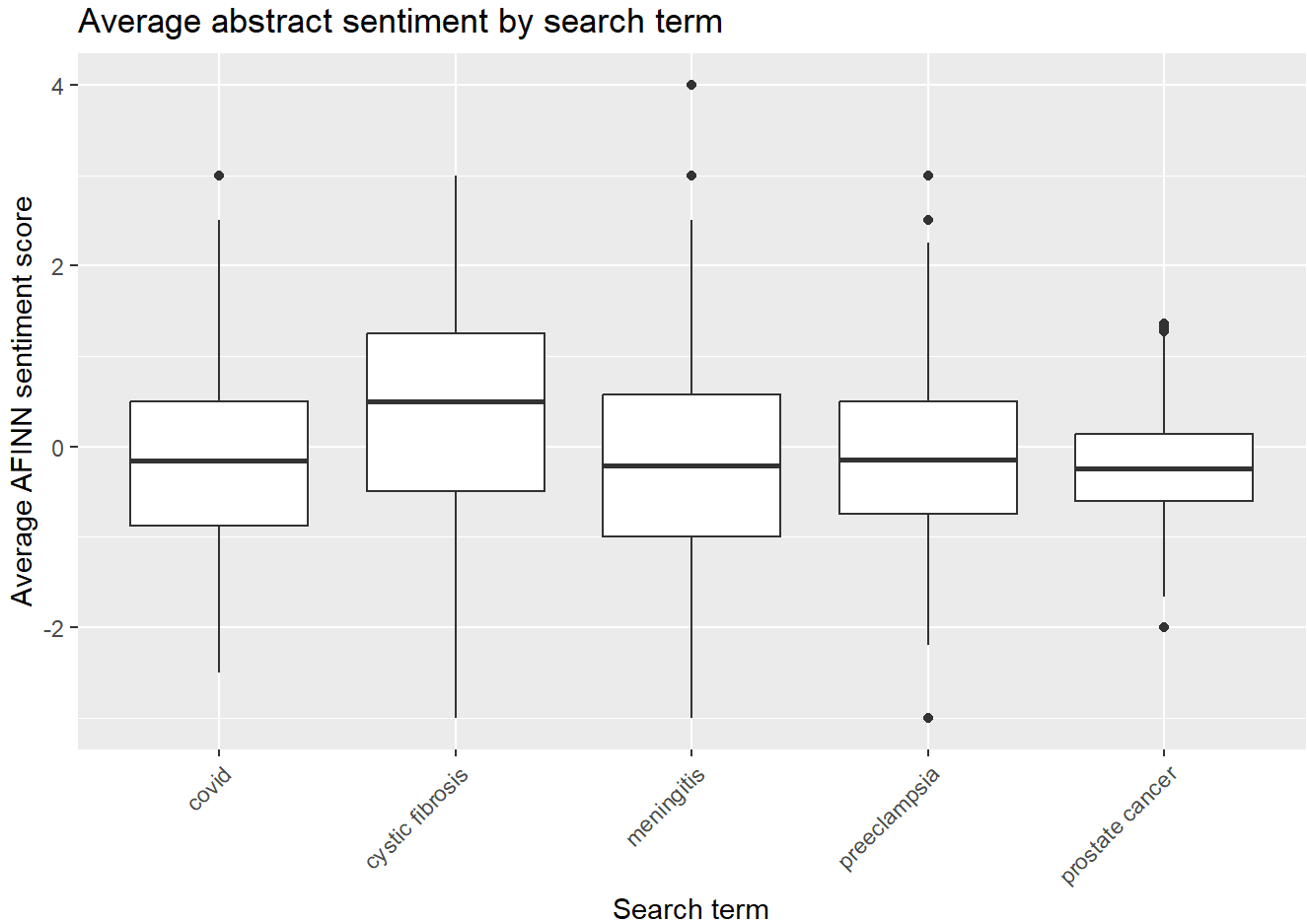
afinn_scores <- pubmed %>%
  unnest_tokens(token, abstract) %>%
  inner_join(afinn, by = c("token" = "word")) %>%
  group_by(term, abstract_id) %>%
  summarize(avg_score = mean(value), .groups = "drop")

head(afinn_scores)
```

```
# A tibble: 6 × 3
  term abstract_id avg_score
<chr>      <int>     <dbl>
1 covid         1    -1.13
2 covid         2     -2
3 covid         3     0.214
4 covid         4    -0.8
5 covid         5    -1.67
6 covid         6    -0.143
```

```
# Visualize average abstract sentiment scores by search term
afinn_scores %>%
  ggplot(aes(x = term, y = avg_score)) +
  geom_boxplot() +
  labs(
    x = "Search term",
    y = "Average AFINN sentiment score",
    title = "Average abstract sentiment by search term"
```

```
) +  
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The search terms do not appear to be dramatically different overall, since most of the boxplots are centered near zero and have substantial overlap. That suggests the average sentiment of the abstracts is fairly similar across topics. The main exception is cystic fibrosis, which has a noticeably higher median AFINN score than the others, indicating its abstracts tend to be somewhat more positive on average. In contrast, covid, meningitis, preeclampsia, and prostate cancer all have medians that are slightly below zero or close to neutral, with fairly similar spreads. There are also outliers for several terms, showing that some individual abstracts are much more positive or negative than the typical abstract, but these extreme values do not seem to change the overall pattern very much.